Non-Invasive Brain Stimulation NIBS

Transcranial Magnetic Stimulation

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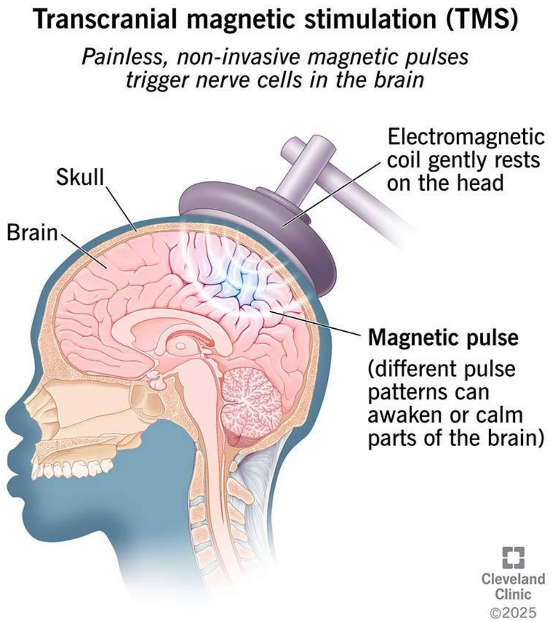
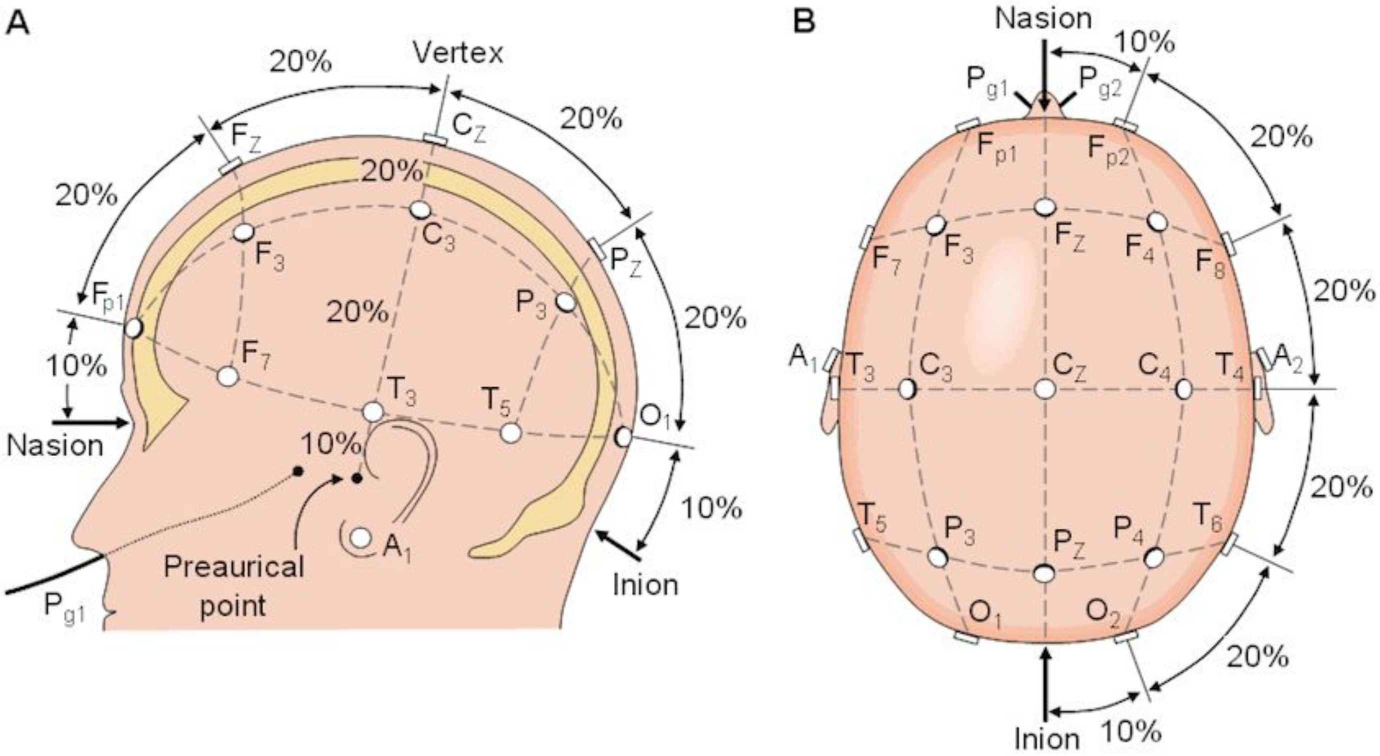
14/06/2026

TMS


Transcranial magnetic stimulation (TMS)
Painless, non-invasive magnetic pulses trigger nerve cells in the brain
Electromagnetic coil gently rests on the head
Skull
Brain
Magnetic pulse
(different pulse patterns can awaken or calm parts of the brain)
Cleveland Clinic
©2025


A
20%
Vertex
Cz
Fz
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20%
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10%
F7
T3
T5
O1
Nasion
10%
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A1
Preaurical
point
Inion
Pg1
B
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10%
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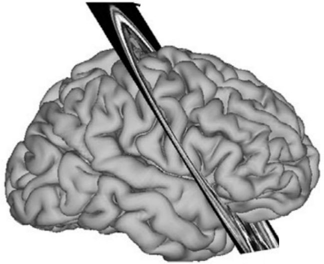
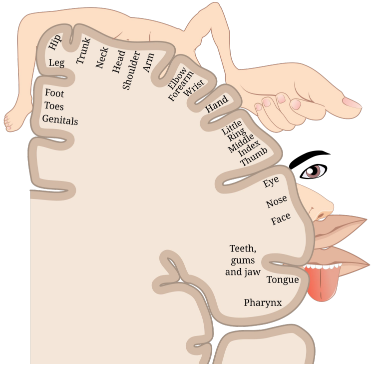
Fp1
Fp2
20%
FZ
F3
F4
F7
F8
20%
A1
T3
C3
CZ
C4
T4
A2
20%
T5
P3
PZ
P4
T6
O1
O2
20%
Inion
10%

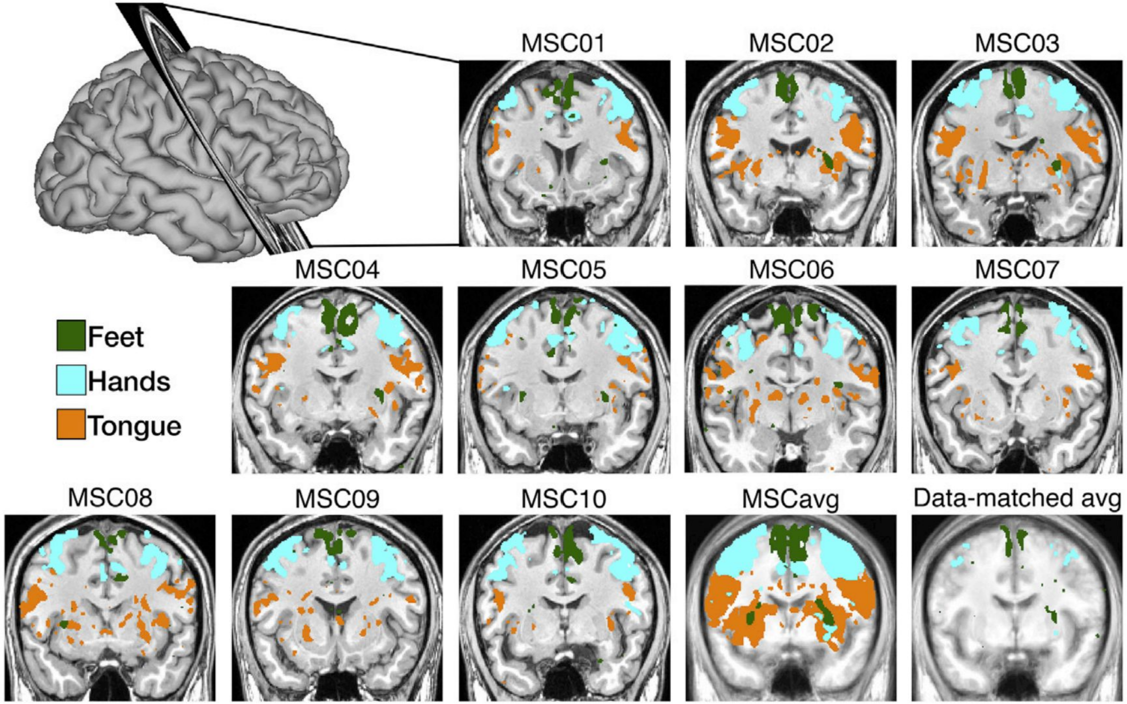
Brain Anatomy and Function varies among people

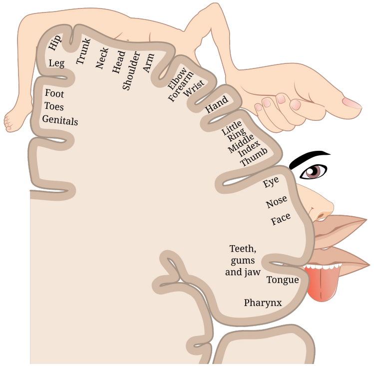
Organización funcional cerebral: la anatomía y la funcion varia entre personas





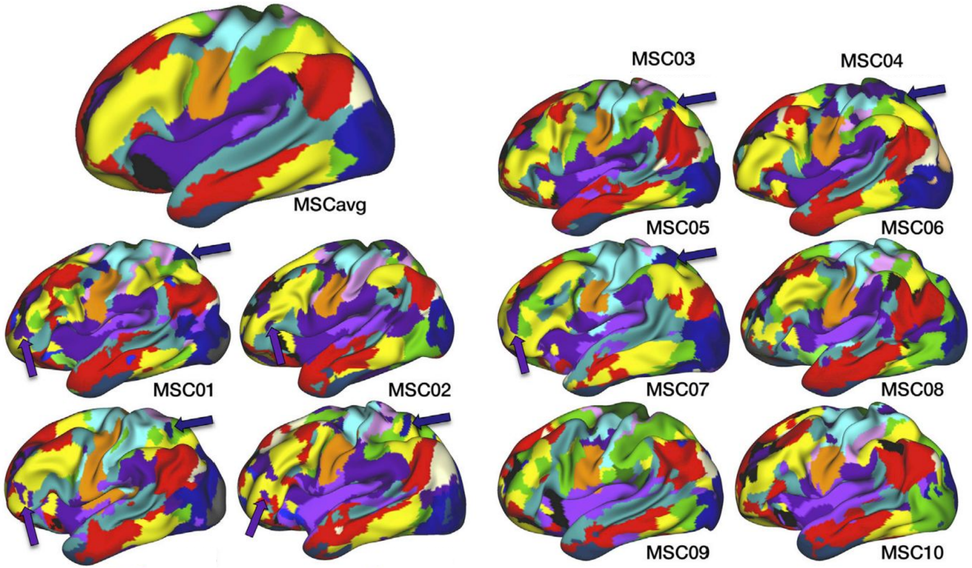
But brain anatomy and functional brain organization varies among people





Gordon, E. M., Laumann, T. O., Gilmore, A. W., Newbold, D. J., Greene, D. J., Berg, J. J., ... & Dosenbach, N. U. (2017). Precision functional mapping of individual human brains. *Neuron*, *95*(4), 791-807.

Functional networks organization varies among individuals on heavily sampled group



- Auditory (Aud)
- Dorsal Attention (DAN)
- Cingulo Opercular (CO)
- Somatomor Dorsal (Smd)
- Temporal pole (Tpole)
- Somatomotor Lateral (SML)
- Visual (Vis)
- Default mode (DMN)
- Medio Temporal (MTL)
- Fronto Parietal (FP)
- Ventral Attention (VAN)
- Parietal Memory (PMN)
- Saliency (Sal)
- Parietal Occipital (PON)

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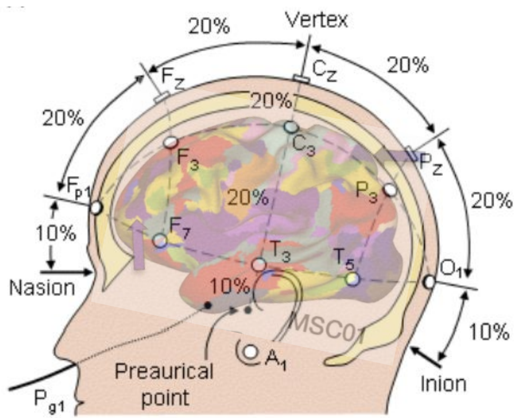
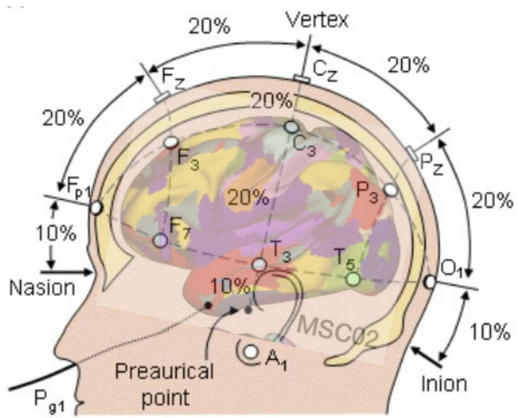
Likely scenario in a TMS clinical intervention

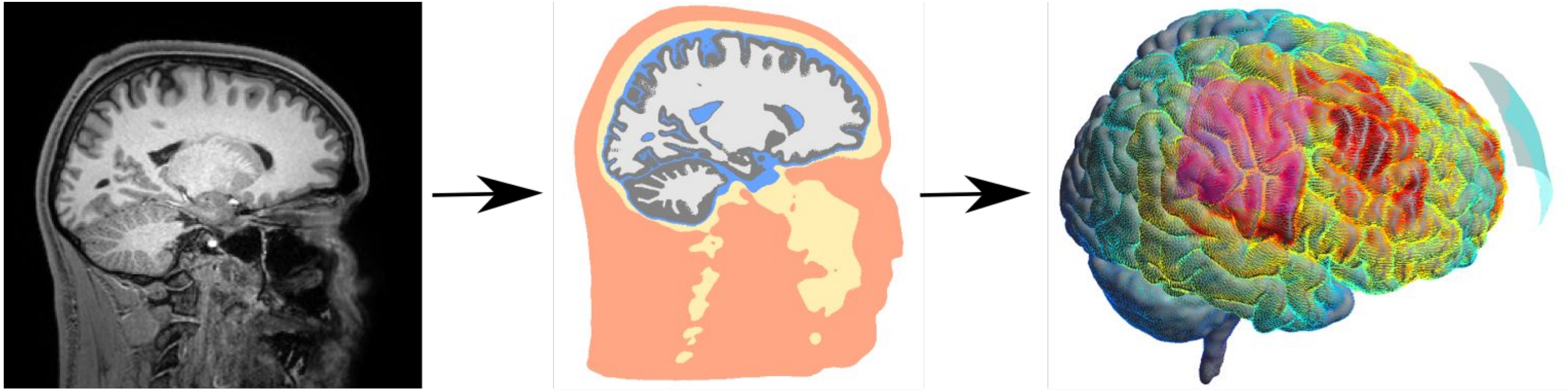
10-20 from EEG does not target same brain netwoks





TMS computational simulation (SimNIBS)

- Finite Element modeling of Magnetic field propagation and Electric field induction

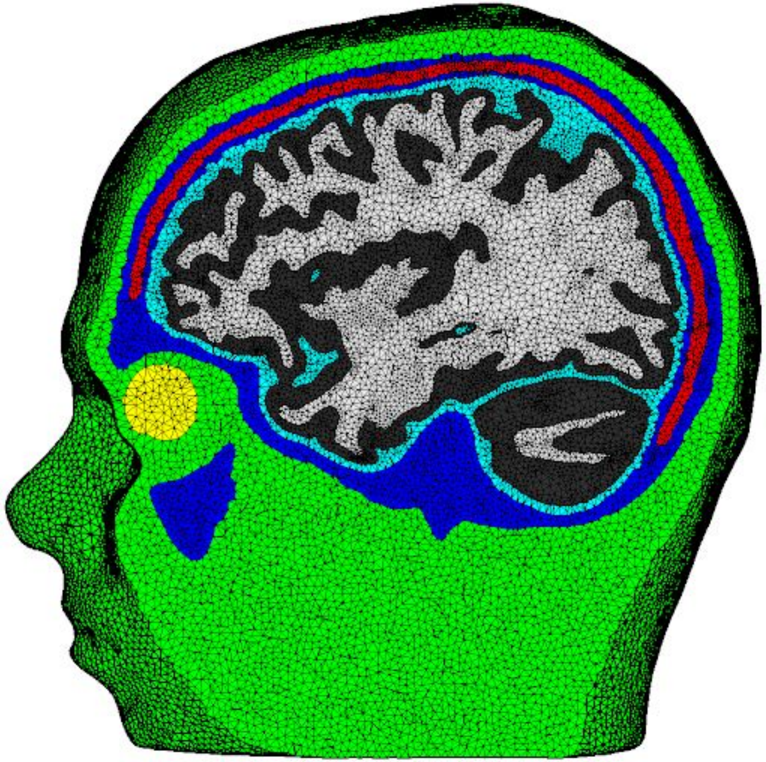


Saturnino, G. B., Puonti, O., Nielsen, J. D., Antonenko, D., Madsen, K. H., & Thielscher, A. (2019). SimNIBS 2.1: a comprehensive pipeline for individualized electric field modelling for transcranial brain stimulation. *Brain and human body modeling: computational human modeling at EMBC 2018*, 3-25.

https://github.com/simnibs/simnibs

Finite Element Models

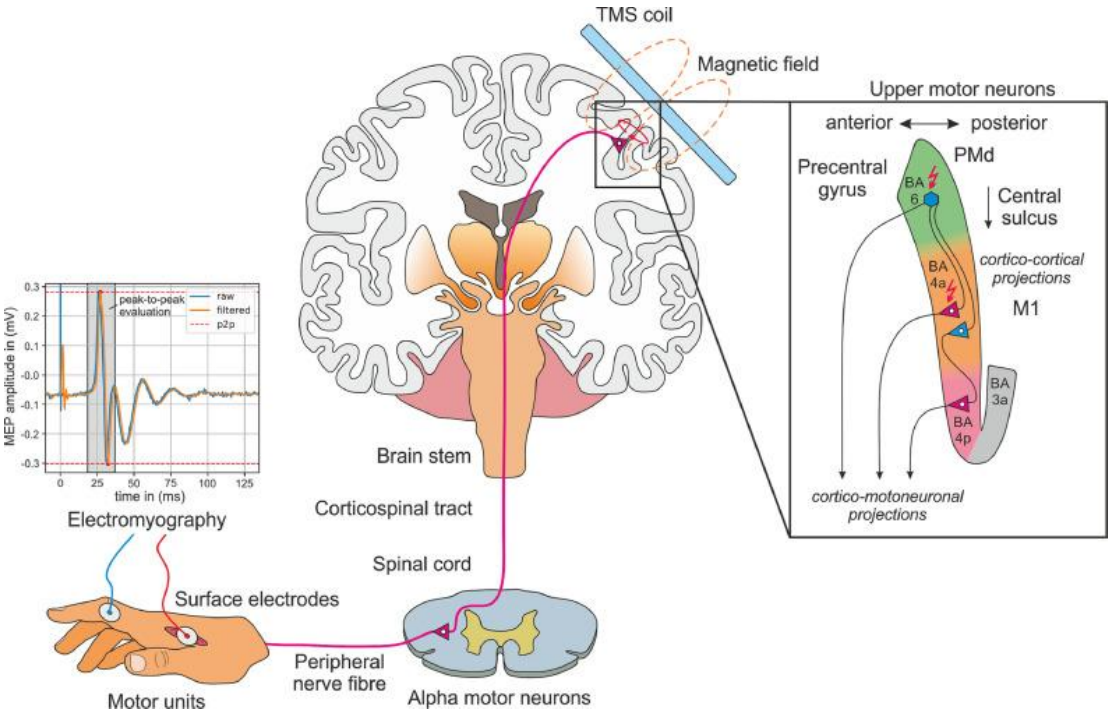
- Each tissue is segmented and meshed
- Differential equations for Efield propagation are solved across tissue. Each tissue has a specific conductivity



http://www.childbrain.eu/childbrain/keski-oikea/esrprojects/esr-13-development-of-new-finite-element-approaches-for-child-brain-research-and-comparison-to-standard-forward-modelling-methods-for-eeg-and-meg-source-analysis

Motor Hotspot

Used for Dose estimation



Numssen, Ole, et al. "Efficient high-resolution TMS mapping of the human motor cortex by nonlinear regression." *bioRxiv* (2021).